

In one line: **Exclusivity** is the rule (no overlapping access where one is a *write*); **ownership** (borrowing/consuming/*~Copyable*) is the vocabulary for who owns a value and when it dies; **layout** (MemoryLayout, existentials, refcount headers) is what all of it costs in bytes and atomic ops.

How it works

- **Law of Exclusivity:** two accesses to the same storage may not overlap in time if at least one *writes*. Reads overlap freely. A mutating call or inout argument is *one* write access lasting the **whole call**.
- **Static check** (local vars, provable aliasing) = compile error, zero cost. **Dynamic** check (class properties, globals, vars captured by escaping closures) = runtime begin/end markers, trap “Simultaneous accesses”. On in **Debug and Release** since Swift 5; `-enforce-exclusivity=unchecked` turns the trap into UB. It is *not* a thread-race detector.
- **~Copyable** (5.9) *suppresses* the implicit Copyable — not a protocol. Structs/enums only (not classes or actors). Values are *moved or borrowed*, never copied $\Rightarrow$ single owner $\Rightarrow$ a **struct can have deininit**, run exactly once when the owner ends or is consumed.
- **Parameter ownership:** borrowing = `+0` / guaranteed, read-only, caller keeps it (default for most params); consuming = `+1` / owned, caller’s binding is dead after; inout = exclusive write, handed back. consume `x` ends `x`’s lifetime explicitly; consuming `func close()` consumes `self`; `discard self` ends it *without* `deininit`.
- Generics: `<T: ~Copyable>` widens (6.0). Optional, `UnsafePointer` accept noncopyables; `Array` still requires `Copyable`.
- **MemoryLayout<T>**: `size` = bytes used; `alignment` = start boundary; `stride` = size rounded up to `alignment` = distance between array elements. Swift keeps **declaration order**, so field order sets padding.
- **Existential any** `P` = 5 words (40 B): 3-word inline buffer + type metadata + 1 witness table per protocol (any `P` & `Q` = 48 B; Any = 32 B; class-bound = `ref` + PWTs). Value > 3 words $\rightarrow$ **boxed on the heap**, buffer holds the pointer. Generics / some `P` specialise and skip the box.
- **ARC header** = 16 B: metadata (isa) + one 64-bit **inline** refcount word (strong + unowned counts + flags). A weak ref or overflow moves counts to a **side table**; the word then points to it. `swift_retain/release` are **atomic** RMWs — cheap once, costly as traffic, worse under contention. `borrowing/+0` and the optimiser remove pairs.
- **Unmanaged<T>**: manual ARC across a C `void*` context. `passRetained(+1)  $\leftrightarrow$  takeRetainedValue(-1)`; `passUnretained  $\leftrightarrow$  takeUnretainedValue(+0)`. `toOpaque/fromOpaque`.

Example

```

struct FileHandle: ~Copyable {
    let fd: Int32
    deinit { close(fd) } // runs exactly once
    consuming func detach() -> Int32 {
        let f = fd; discard self; return f } // no deinit
}

func read(_ h: borrowing FileHandle) {} // lend, +0
func store(_ h: consuming FileHandle) {} // take, +1
let h = FileHandle(fd: 3)
read(h); store(h)
// read(h) error: 'h' used after consume
var x = 1
swap(&x, &x) // error: overlapping accesses to 'x'

```

Padding: same fields, new order

struct A { a: Bool; b: Int64; c: Bool }

size 17 (7 B inner padding after a)

stride 24 - alignment 8

inner pad tail pad

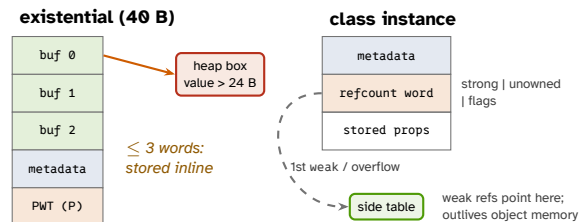
struct B { b: Int64; a: Bool; c: Bool }

size 10

stride 16 - alignment 8

*biggest alignment first:
saves 8 B per element*

any P and the object header



Traps

- ! Array indexing / pointer maths uses **stride**, not size (Int?: size 9, stride 16).
- ! Exclusivity $\neq$ data-race safety — that is actors / Sendable. It catches single-threaded reentrancy.
- ! `&a[0]`, `&a[1]` conflict: subscript in/out accesses the *whole array*. Use `swapAt`.
- ! `~Copyable` is suppression, not conformance; a borrowing param cannot be returned or stored (escapes the borrow).
- ! any `P` in a hot loop may `malloc` (box) and adds dynamic dispatch — prefer some `P` / generics.
- ! `passRetained + takeUnretainedValue = leak`; the reverse = over-release crash.
- ! “Value type = no ARC” — a struct holding a class ref retains it on copy.

Remember

“One writer, whole call.” Borrow reads, Consume takes, Inout rents — “BCI”. Pack fields **big-to-small**. **3 words** fit in an existential; **weak** buys a side table.